

Task 1 - network topology

Assumptions

- At max there are 4 adjacent EV charging stations, in adjacent , diagonal charging stations are not linked.

Required comparisons for task 2

- Node degree (maximum 4, minimum 2) and all details from the lecture given table

Extras

- Scalability
-

Properties that have to be optimized for the EV charging nodes

Latency refers to the total time taken

Bandwidth refers to the total amount of data that can be carried in a pipeline at a given time

- Latency and bandwidth
 - Other EV charging nodes
 - Base station

Task 1 - network topology

<i>Topology</i>	<i>Node degree</i>	<i>Diameter</i>	<i>Bisection width</i>	<i>Arc connectivity</i>	<i>Cost</i>
Linear array	1 or 2	$N - 1$	1	1	$N - 1$
Ring	2	$N/2$	2	2	N
Star	1 or $N - 1$	2	1	1	$N - 1$
Binary tree	1, 2 or 3	$2\log((N+1)/2)$	1	1	$N - 1$
2-D mesh	2, 3 or 4	$2(N^{1/2}-1)$	$N^{1/2}$	4	$2(N-N^{1/2})$
2-D wraparound	4	$N^{1/2}$	$2N^{1/2}$	4	$2N$
3-D cube	3, 4, 5 or 6	$3(N^{1/3}-1)$	$N^{2/3}$	3	$2(N-N^{2/3})$
Hypercube	$\log N$	$\log N$	$N/2$	$\log N$	$(N\log N)/2$
Completely connected	$N - 1$	1	$N^2/4$	$N - 1$	$N(N-1)/2$

Add reference to thi image

Task 1 - network topology

Topology	Scalability	Fault tolerance	WSN geographic realism	Immediate-neighbour communication	Main weakness for this task	Suitability
Linear array	Moderate	Poor	Poor	Limited to 2 neighbours	Long diameter; unrealistic for a 2-D area	Low
Ring	Moderate	Better than linear	Poor	Only 2 neighbours	Artificial wraparound/loop does not represent geographic adjacency	Low
Star	Moderate	Poor central failure tolerance	Moderate	Poor	Charging nodes communicate mainly through centre rather than neighbouring stations	Low–Medium
Binary tree	Good	Poor around upper-level links	Poor	Poor	Hierarchical links do not represent nearby physical charging stations	Low
2-D mesh	Good	Good	Excellent	Excellent	Diameter grows as network grows	Best
2-D wraparound / torus	Good	Very good	Moderate	Good	Edge nodes connecting to opposite side is geographically unrealistic	Medium–High
3-D cube	Good	Good	Poor	Poor–Moderate	Adds a third spatial dimension not required by the road/grid model	Medium
Hypercube	Excellent	Excellent	Poor	Poor	Logical neighbours may be physically very far apart	Medium
Completely connected	Poor hardware scalability	Excellent	Poor	Excessive	Requires links; every station communicates directly with every other	Low

Task 1 - network topology

Geographic realism

EV charging stations are physically distributed over a two-dimensional geographical area.

A 2-D mesh naturally models this.

A hypercube may have better mathematical communication properties, but its neighbours are determined logically rather than by geographical proximity.

That makes the mesh a more realistic simulation model.

Immediate-neighbour communication

The specification specifically says charging nodes communicate with immediate adjacent nodes.

That is fundamentally what mesh connectivity represents.

This is probably your single strongest argument.